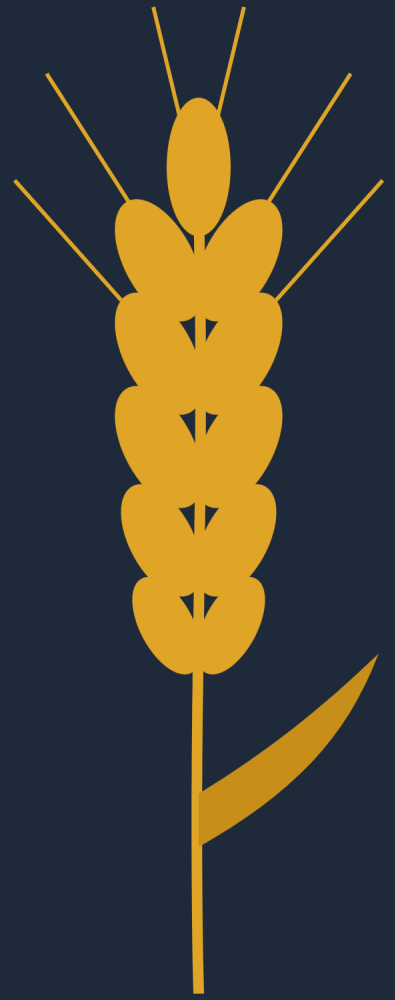
ICONIC CLUB AFACERI.RO · IAȘI

Financial Instruments

Explained with one bushel of wheat

Iulian Costan



OVERVIEW

What we'll cover tonight

PART ONE

Spot (cash) market

- **Spot instruments** · own it today
- **Pure arbitrage** · Chicago → Detroit

PART TWO

Derivatives

- **CFD** · leverage and margin
- **Futures** · farm, mill, basis
- **Options** · floor, ceiling, strike
- **Statistical arbitrage**

BEFORE WE START

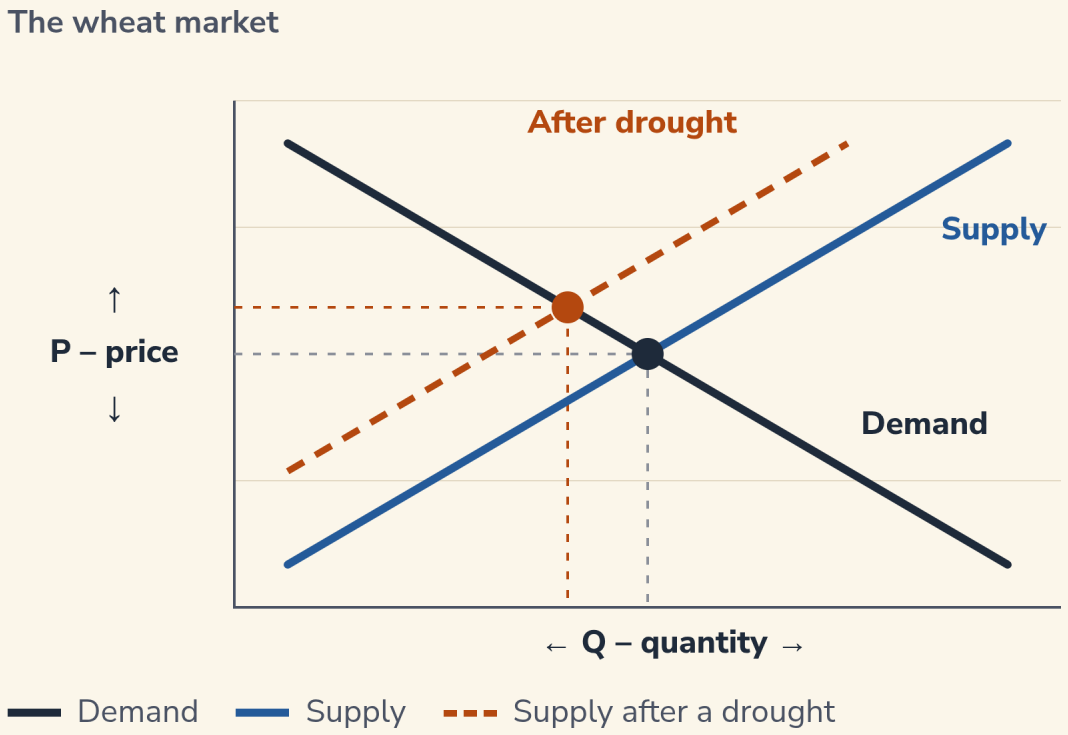
Where does a price come from?

Demand (mills, bakeries): they buy more when wheat is cheap.

Supply (farms): they sell more when wheat is expensive.

The price settles where the two meet: **$100 a bushel**.

⚠ **A drought cuts supply:** the price climbs to about $106.



PART ONE

Spot (cash) market

You pay the full price, and the thing is yours, today.

BID $99

The best price a buyer will **pay you**. You sell at the bid.

SPREAD $2

Ask minus bid: the **cost** of trading right now.

ASK $101

The best price a seller will **take from you**. You buy at the ask.

Wheat at the silo, or stocks, bonds and ETFs on the exchange.

1 · SPOT MARKET

Pay now, get it now

At the silo in Chicago

You pay $100 and get a bushel of wheat. On the spot. It's yours, and you take it home.

On the exchange

You pay $100 and get stocks, bonds or ETFs. They land in your account, and they're yours.

Spot means you own it. Settlement on an exchange takes a day or two, but it's still "now".

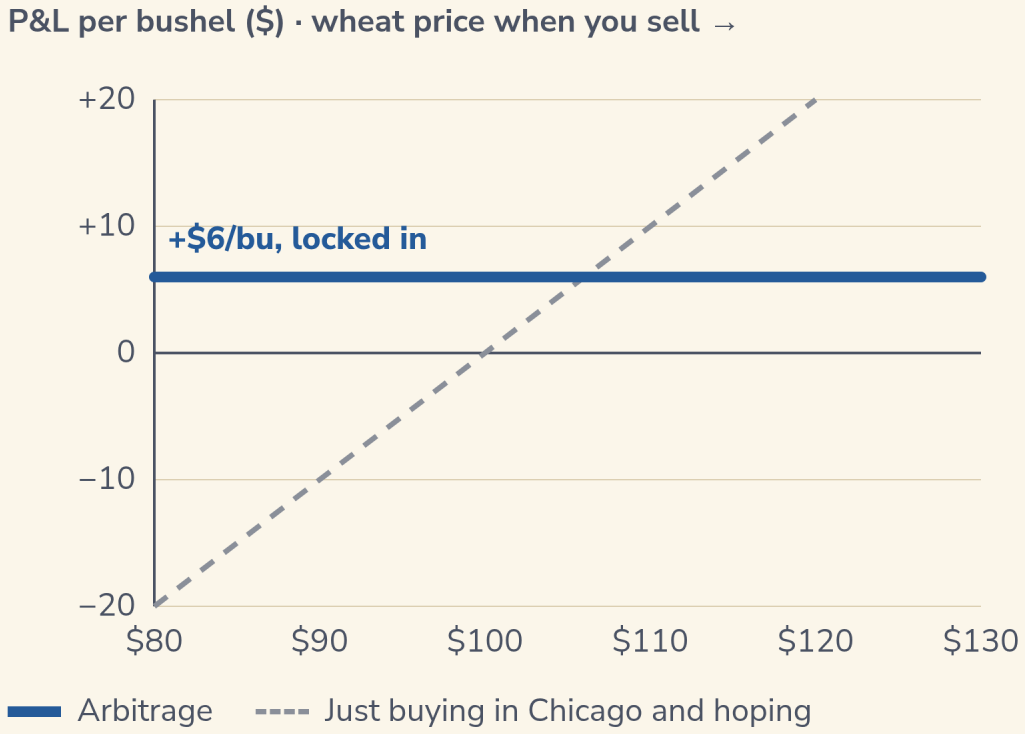
1 · PURE ARBITRAGE

Same wheat, two prices

Per bushel	$
Buy in Chicago	100
+ Transport	4
+ Kiosk and fees	2
Sell in Detroit	112

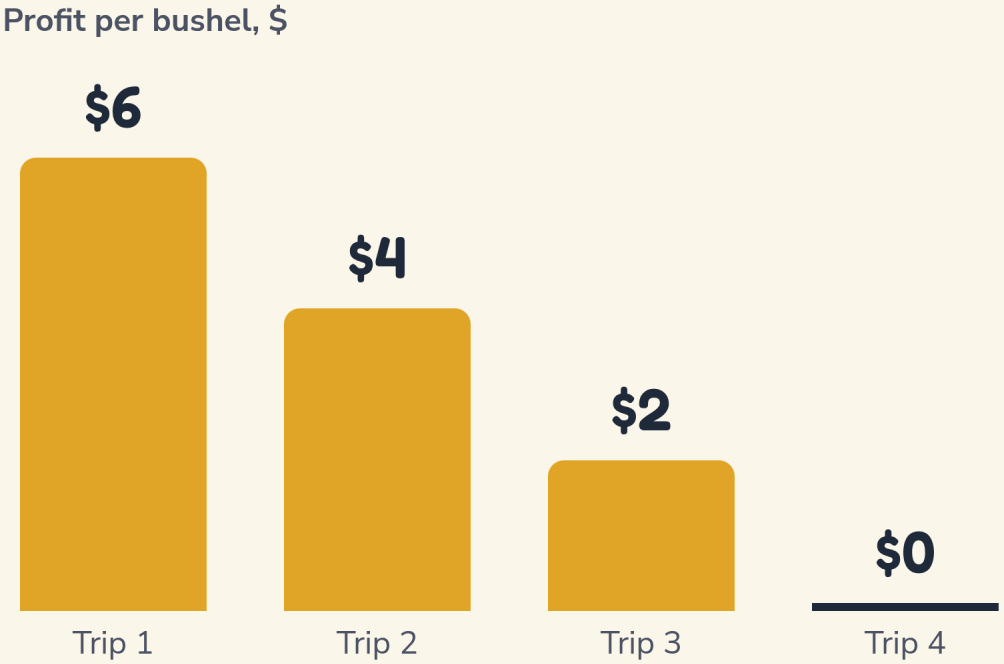
Profit: $6 per bushel

Buy and sell at the same moment. No forecast needed.



1 · PURE ARBITRAGE

Every truckload shrinks the edge



We buy in Chicago, so demand there grows and the price **rises**. We sell in Detroit, so supply there grows and the price **falls**.

The gap ends up covering only transport and the kiosk. The market is back in **equilibrium**.

And someone faster usually spots the gap first.

PART TWO

Derivatives

You don't buy the wheat. You buy a contract on the price of wheat.

LONG ↑

You **buy** the contract. You profit when the price **rises**.

SHORT ↓

You **sell** the contract first. You profit when the price **falls**.

The contract's value "derives" from something else: wheat, oil, stocks, indices, currencies.

What changes compared with spot

	Spot	Derivatives
What you buy	The goods or the security	A contract on the price
What you pay	The full price	A deposit (margin)
You profit when…	The price rises	It rises or falls, your choice
Duration	As long as you like	Often has an expiry
Risk	At most what you invested	The margin can vanish fast

2 · LEVERAGE AND MARGIN

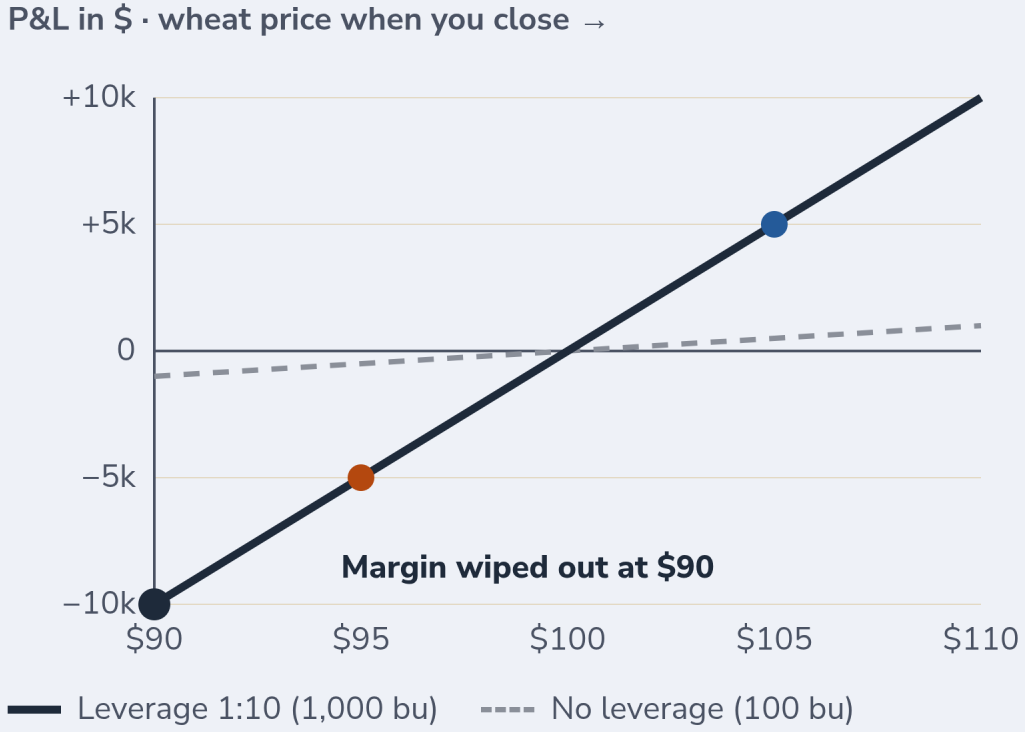
$10,000 that acts like $100,000

Margin $10,000 × leverage 1:10 = exposure $100,000, i.e. 1,000 bushels at $100.

WHEAT +5%
Gain: +$5,000
+50%

WHEAT −5%
Loss: −$5,000
−50%

WHEAT −10%
Margin call: position closed
$0



2.1 · CFD

CFD: a bet on the price difference

Contract for Difference. You and the broker settle the gap between entry and exit price.

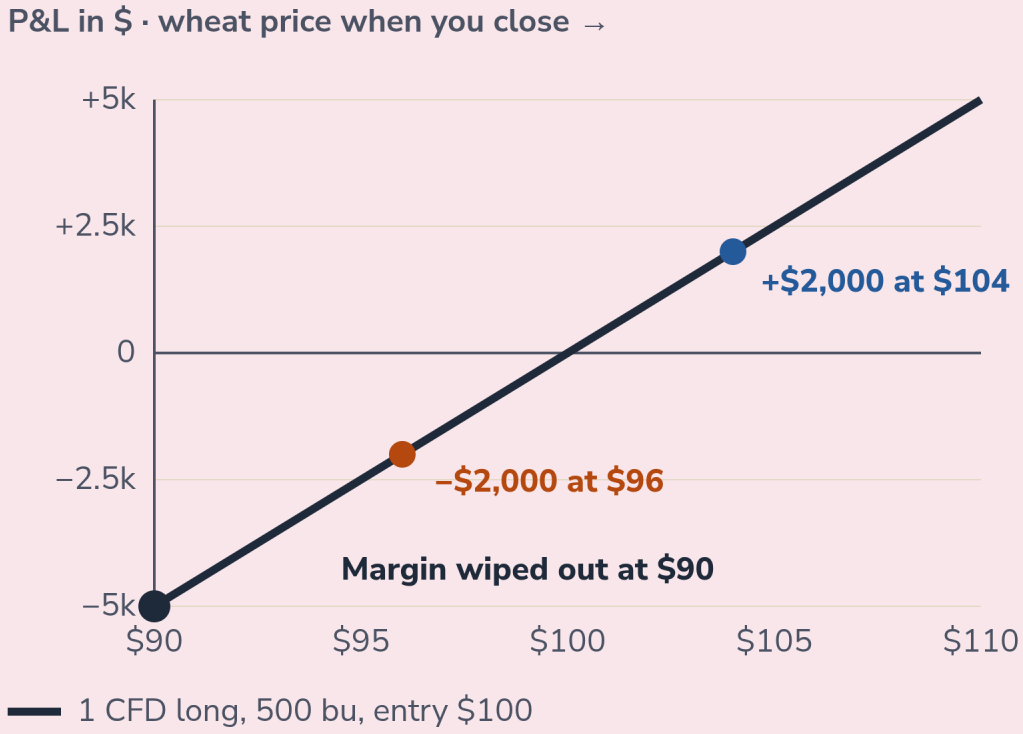
Example: wheat CFD

Contract size: **500 bushels**

At $100/bu → exposure **$50,000**

10% margin: **$5,000**

Unlike spot: you never own the wheat, you pay financing every night, and the broker is your counterparty.



2.2 · FUTURES – INTRODUCTION

Futures: a price fixed today, with a delivery date

Real example: CBOT wheat in Chicago, 5,000 bushels per contract, expiries in March, May, July, September and December.

	CFD	Futures
Where	With a broker	On an exchange, backed by a clearing house
Expiry	None	A fixed date
Delivery	Never	Possible, physically
Who uses it	Mostly speculators	Farms, mills, traders, funds

2.2 · FUTURES – FARM EXAMPLE

The farm locks in the price in March

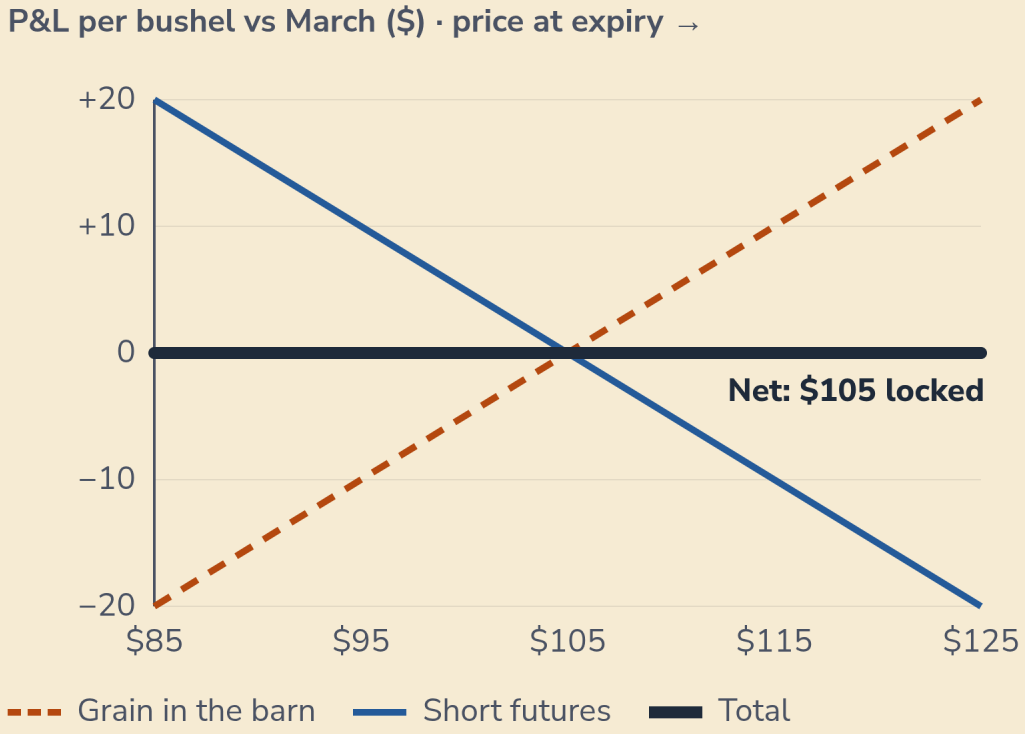
March spot: **$110**

Sep futures: **$105**

Its crop is still in the field, so it can't sell spot. It **sells (goes short) September futures at $105.**

September	Grain sale	Futures	Net /bu
Falls to $90	$90	+$15	$105
Rises to $120	$120	–$15	$105

$105/bu either way. Easier credit, better planning.



2.2 · FUTURES – MILL EXAMPLE

The mill locks in the cost of flour

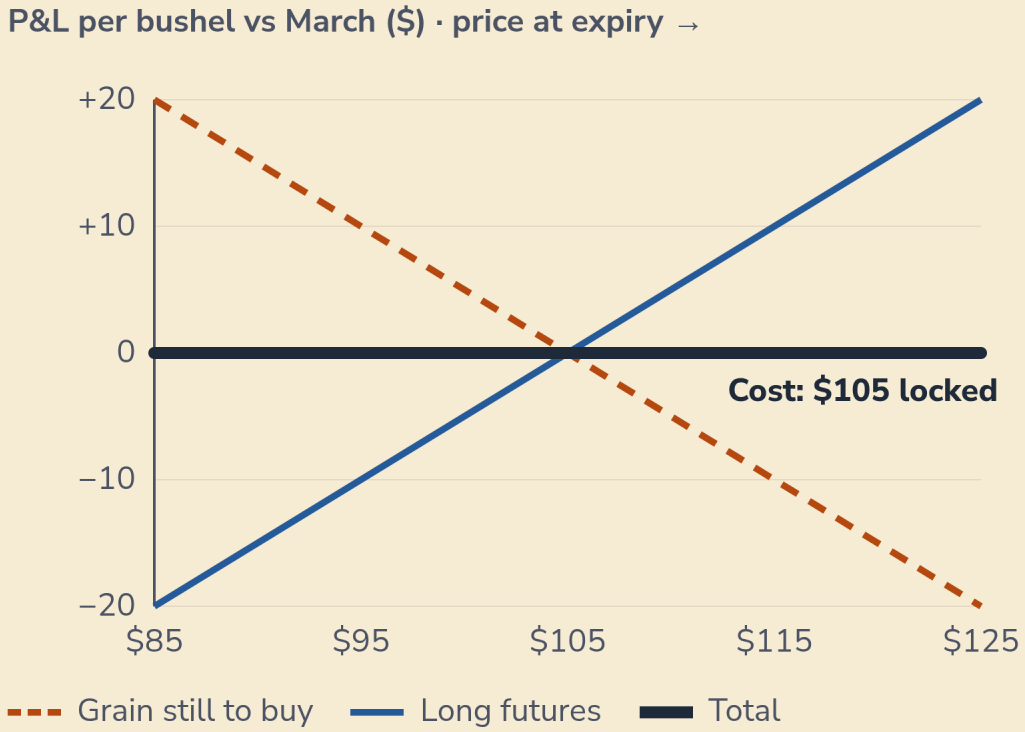
March spot: **$110**

Sep futures: **$105**

It needs wheat in September, not now, and fears a drought. It **buys (goes long) September futures at $105.**

September	Grain cost	Futures	Net /bu
Rises to $120	$120	+$15	$105
Falls to $90	$90	–$15	$105

A fixed cost: it can sign year-long contracts with bakeries.



2.2 · FUTURES – BASIS

Basis: spot minus futures

July: harvest

Wheat everywhere, silos are full, storage is expensive.

Spot **$90** · December futures **$98**

Basis –$8: you pay to wait

What it means: futures sit above spot by the cost of storage. The market pays whoever stores wheat until December.

March: before harvest

Silos are almost empty and the mills are hunting for wheat.

Spot **$110** · September futures **$105**

Basis +$5: wheat today is scarce

What it means: spot sits above futures because wheat is short now. The market pays whoever sells from the silo today.

At expiry the basis goes to zero: futures and spot meet.

Options: the right, not the obligation

CALL

The right to **buy** at a price fixed today. It puts a ceiling on your cost.

PUT

The right to **sell** at a price fixed today. It puts a floor under your price.

You pay a **premium**, like insurance. If you don't need it, you let it expire and lose only the premium.

2.3 · OPTIONS – FARM EXAMPLE

The farm buys a floor

March spot: **$110**

Sep futures: **$105**

Crop still in the field: no spot sale in March. **$105 in September is fine; below $95 the farm is in trouble.**

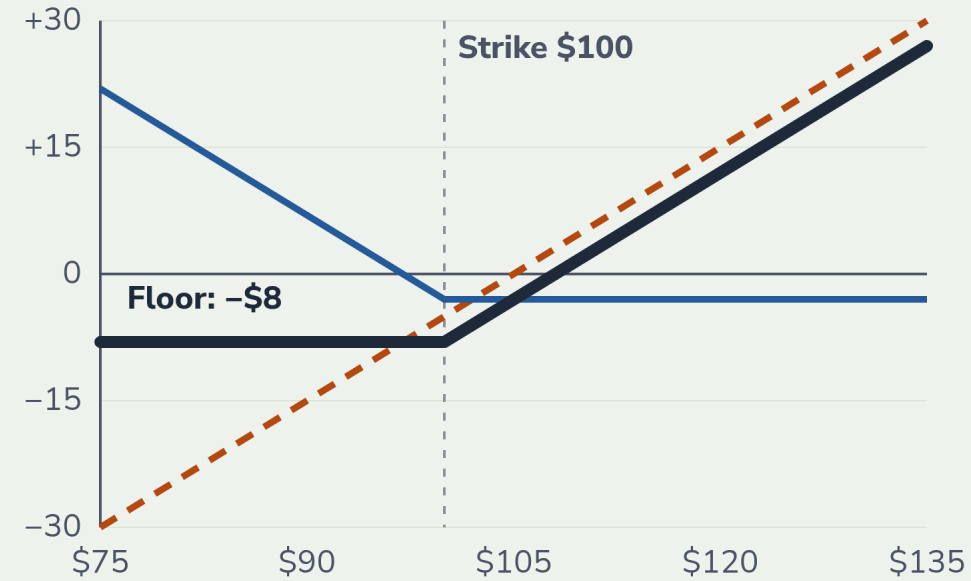
Buys a PUT · strike $100 · premium $3/bu

September	Grain sale	Put	Net /bu
Falls to $80	$80	+$17	$97
Falls to $90	$90	+$7	$97
Rises to $120	$120	–$3	$117

Put column after the $3 premium: +$10 payout – $3 = +$7.

Floor at $97, and the upside stays its own.

P&L per bushel vs Sep futures $105 · price at expiry →



–$8 = strike $5 below futures ($100 vs $105) + $3 premium

2.3 · OPTIONS – MILL EXAMPLE

The mill buys a ceiling

March spot: **$110**

Sep futures: **$105**

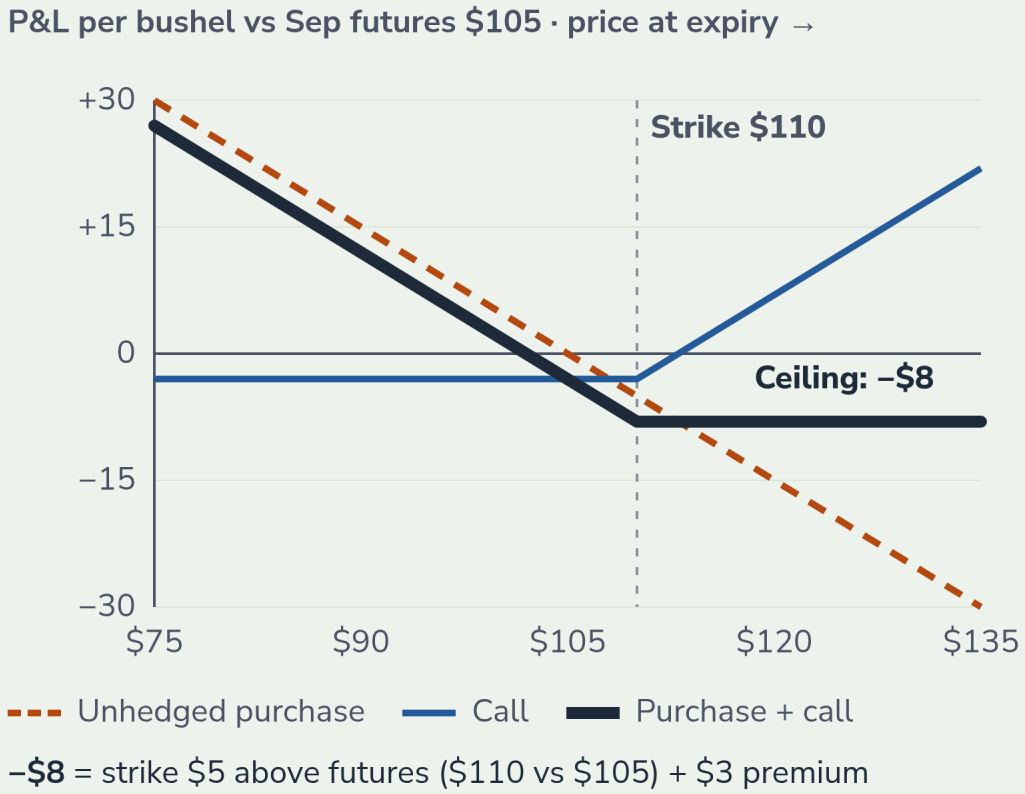
It needs wheat in September, not now. **$105 is fine; above $115 the business is in danger.**

Buys a CALL · strike $110 · premium $3/bu

September	Grain cost	Call	Net /bu
Rises to $130	$130	+$17	$113
Rises to $120	$120	+$7	$113
Falls to $90	$90	–$3	$93

Call column after the $3 premium: +$10 payout – $3 = +$7.

Ceiling at $113, and cheaper wheat stays its gain.



–$8 = strike $5 above futures ($110 vs $105) + $3 premium

2.3 · OPTIONS – STRIKE PRICE

Strike: where the option starts to pay

Wheat at expiry	Farm's PUT, strike $100	Mill's CALL, strike $110
$90 (big crop)	In the money: worth $10	Out of the money: $0
$105 (normal year)	Out of the money: $0	Out of the money: $0
$120 (drought)	Out of the money: $0	In the money: worth $10

STRIKE

The fixed price written in the option: the price you may sell at (put) or buy at (call).

SPOT VS STRIKE

Decides the value at expiry: in the money it pays, out of the money it's worth $0.

LIKE BASIS FOR FUTURES

A better strike (higher floor, lower ceiling) costs a higher premium.

You win on average over a large number of trades

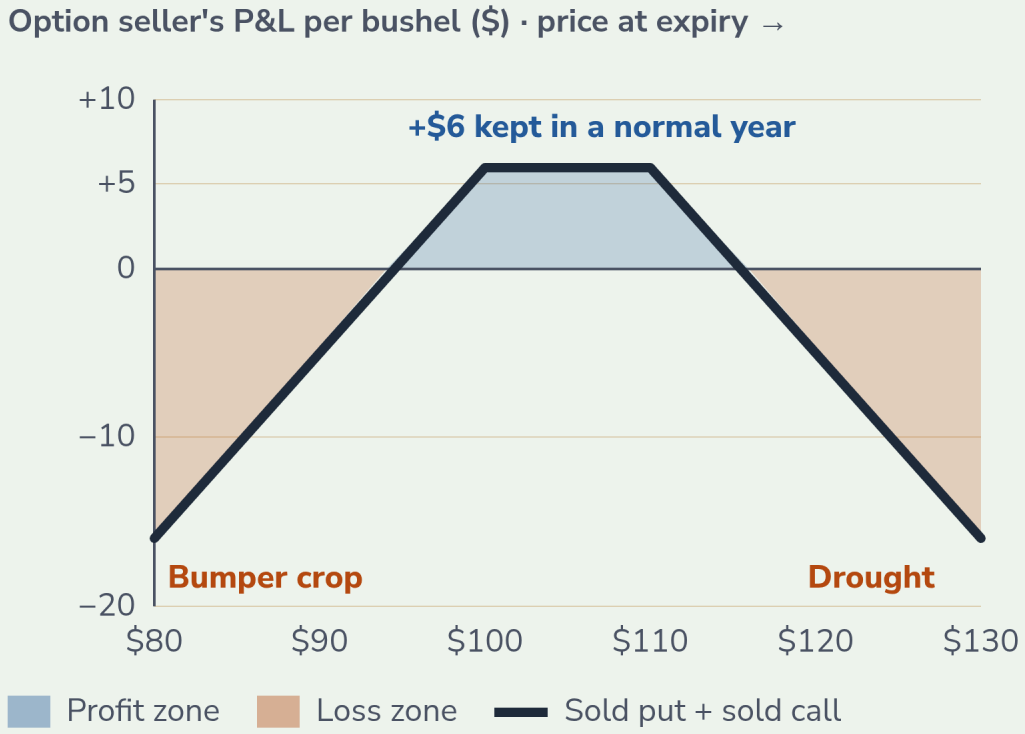
PURE ARBITRAGE

High edge (high expected value, high probability of profit) per trade.

STATISTICAL ARBITRAGE

A small edge, repeated thousands of times.

Who sells the farm's put and the mill's call? A trader acting like an **insurer**: keeps $6/bu in a normal year, pays out in extreme ones.



RECAP

Four instruments on one page

Instrument	What you buy	Leverage	Expiry	Good for
Spot	Ownership or a claim	No	Only bonds	Long-term investing
CFD	The price difference, with a broker	High	No	Short-term speculation
Futures	A commitment, on an exchange	Yes	Yes	Locking in a price (hedging)
Options	A right, paid with a premium	Yes	Yes	Insurance: a floor or a ceiling

ASSET CLASSES · ONE EXAMPLE ASSET EACH

Bonds
UST
US Treasuries

FX
EURUSD
Euro vs. dollar

Stocks
TSLA
Tesla

Commodities
ZW
Wheat (CBOT)

Crypto
BTC
Bitcoin

Three things to remember

1. Spot means you own it. A derivative is a contract on the price.
2. Leverage magnifies everything: gains and mistakes.
3. Hedging isn't a bet. It's insurance for the business.

Questions?

Educational material, not investment advice. Prices in the examples are illustrative.